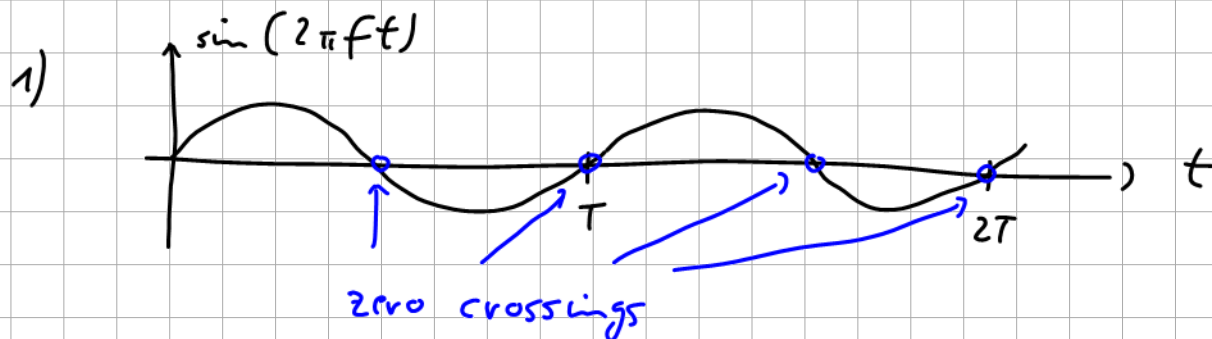




Obviously one period T of a sinus has 2 zero crossings.

$$\Rightarrow f = 1234 \text{ Hz} \quad \Rightarrow \quad 2468 \text{ zero crossing \# per second}$$

2 zero crossings for one period length

one period length is $T = \frac{1}{f}$ in seconds or
 $\frac{r}{f}$ in samples

$$\Rightarrow zcr = \frac{2f}{r}$$

e.g. for a sampling rate of $r = 16000$

$$zcr = \frac{2 \cdot 1234}{16000}$$

3)



linearity: $\Sigma(ax_1 + bx_2) = a \cdot \Sigma(x_1) + b \cdot \Sigma(x_2)$

In order to prove non-linearity, one counter example for the above given equation is sufficient:

$$\begin{array}{ll} a = 1 & x_1 = 2 \\ b = -1 & x_2 = 1 \end{array}$$

$$\Sigma(1 \cdot 2 + (-1) \cdot 1) = 1 \cdot \Sigma(2) - 1 \cdot \Sigma(1)$$

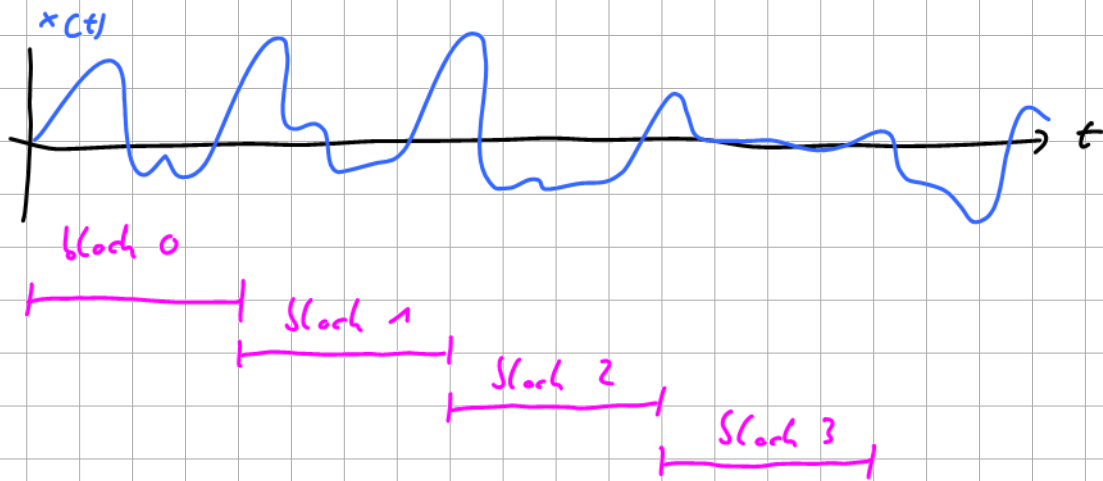
$$\Sigma(1) = 1 - 1$$

$$1 = 0 \quad \Downarrow$$

non-linearity is proven!

4) $\text{numpy.arange}(5) \hat{=} [0 \ 1 \ 2 \ 3 \ 4]$

Assuming a signal $x(t)$ and blocks of a given length in milliseconds:



The features are assumed to be valid for the whole block.

In order to align the signal x_{mid} and the features properly, the feature is assumed to be located in the middle of the block.

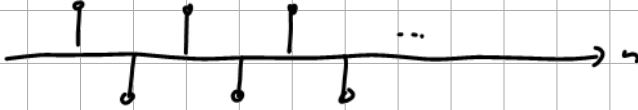
The middle of the block n is:

$(n + 0,5) \cdot T$ with T corresponding to the blocksize in seconds.

5) smallest possible ZCR: no zero crossings in the observed signal:

$$\underline{\underline{ZCR = 0}}$$

highest possible ZCR: each sample changes the sign:



$$\Rightarrow \underline{\underline{ZCR = 1}}$$

$$2) \quad f_0 = (1; 0) \quad c_1 = (5; 0)$$

$$f_1 = (0; 1) \quad c_2 = (0; 5)$$

$$f_2 = (3; 1)$$

$$f_3 = (1; 6)$$

Distance between all feature vectors and all centers:

$$d_{01} = (1-5)^2 + (0-0)^2 = 16 \quad d_{02} = (1-0)^2 + (0-5)^2 = 26$$

→ f_0 belongs to c_1

$$d_{11} = (0-5)^2 + (1-0)^2 = 26 \quad d_{12} = (0-0)^2 + (1-5)^2 = 16$$

→ f_1 belongs to c_2

$$d_{21} = (3-5)^2 + (1-0)^2 = 5 \quad d_{22} = (3-0)^2 + (1-5)^2 = 25$$

→ f_2 belongs to c_1

$$d_{31} = (1-5)^2 + (6-0)^2 = 52 \quad d_{32} = (1-0)^2 + (6-5)^2 = 2$$

f_3 belongs to c_2

Update centers:

$$c_1 = \frac{1}{2} [(1; 0) + (3; 1)] = (2; \frac{1}{2})$$

$$c_2 = \frac{1}{2} [(0; 1) + (1; 6)] = (\frac{1}{2}; \frac{7}{2})$$

→ the new centers are the mean values of their associated feature vectors.

The k-means now starts again with evaluating all distances.





